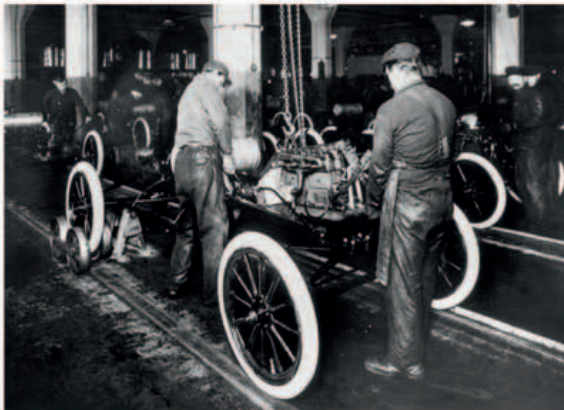


THE STORY OF THE CAR

FROM THE MODEL T TO HYBRIDS, CARS HAVE BEEN A DRIVING FORCE FOR CHANGE

After thousands of years of slow transportation using ships, horses, or even traveling on foot, the development of the automobile revolutionized the way the world thought about distance and speed. Instead of spending days on a trip, people and goods could move hundreds of miles in a matter of hours.

Although the late 19th century witnessed many significant technological innovations in the realm of transportation, such as the development of steamships, none would come close to having the widespread and immediate impact of the development and mass production of the car. Although automobile ownership was at first only the preserve of the wealthy, the US inventor Henry Ford was able to increase output and push down



Rolling off the assembly line

Workers on the assembly line at the Ford Motor Company assemble a Model T. Ford's innovative factories allowed the company to assemble millions of cars very quickly.

prices, so that by the 1920s many eager consumers could buy a car. This had a profound effect on the landscape as highways sprang up, and by the 1950s, suburbs in the US were planned around the idea that residents would be driving.

THE AGE OF THE AUTOMOBILE

Despite the subsequent problems—especially pollution and traffic jams—the love affair with the car has never ceased. Indeed, as people in developing countries become richer, they too want to be car owners. Now the challenge is to find more fuel-efficient and environmentally friendly ways to power cars, and more manufacturers are experimenting with other forms, such as hybrids (see panel, right). However, in spite of these issues, the automobile continues to be an integral part of transportation networks all over the world.



Ford Model T



HYBRID CARS

Automobile engineers have long been trying to find ways to run cars on other fuels than gasoline, including solar power and battery power. Hybrid cars combine a fuel engine with a battery engine, giving the driver better fuel consumption and producing less pollution.

“ ANY CUSTOMER CAN HAVE A CAR PAINTED ANY COLOR THAT HE WANTS SO LONG AS IT'S BLACK. ”

Henry Ford, US industrialist, *My Life and Work*, 1922

15th century
Leonardo da Vinci's car
The Renaissance Italian designs the world's first self-propelled wagon.



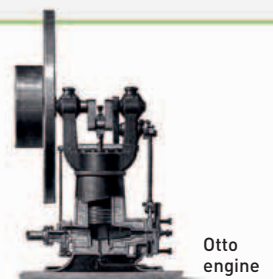
Reconstructed da Vinci car

Trevithick's road locomotive

1801
The steam-powered car
Richard Trevithick, a British inventor, creates a smaller, lighter version of the steam engine and calls it the "road locomotive."



1867–77
The four-stroke Otto engine
The German inventor Nikolaus August Otto patents his four-stroke internal-combustion engine.



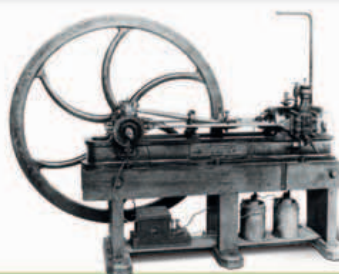
Otto engine

1769–70
The first true automobile
The French engineer Nicolas Cugnot builds a steam-powered vehicle that can reach speeds of up to 2 mph (3 kph).



Cugnot's Faradier

1860
The coal-gas engine
Belgian Jean-Joseph Étienne Lenoir invents a two-stroke internal-combustion engine fired by coal gas.



Lenoir gas engine

Benz three-wheeler



1885
Internal combustion improves
The Germans Karl Benz and Gottlieb Daimler separately develop practical cars with internal-combustion engines.